RPMGLOBAL

AMT - System Administration

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. This guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

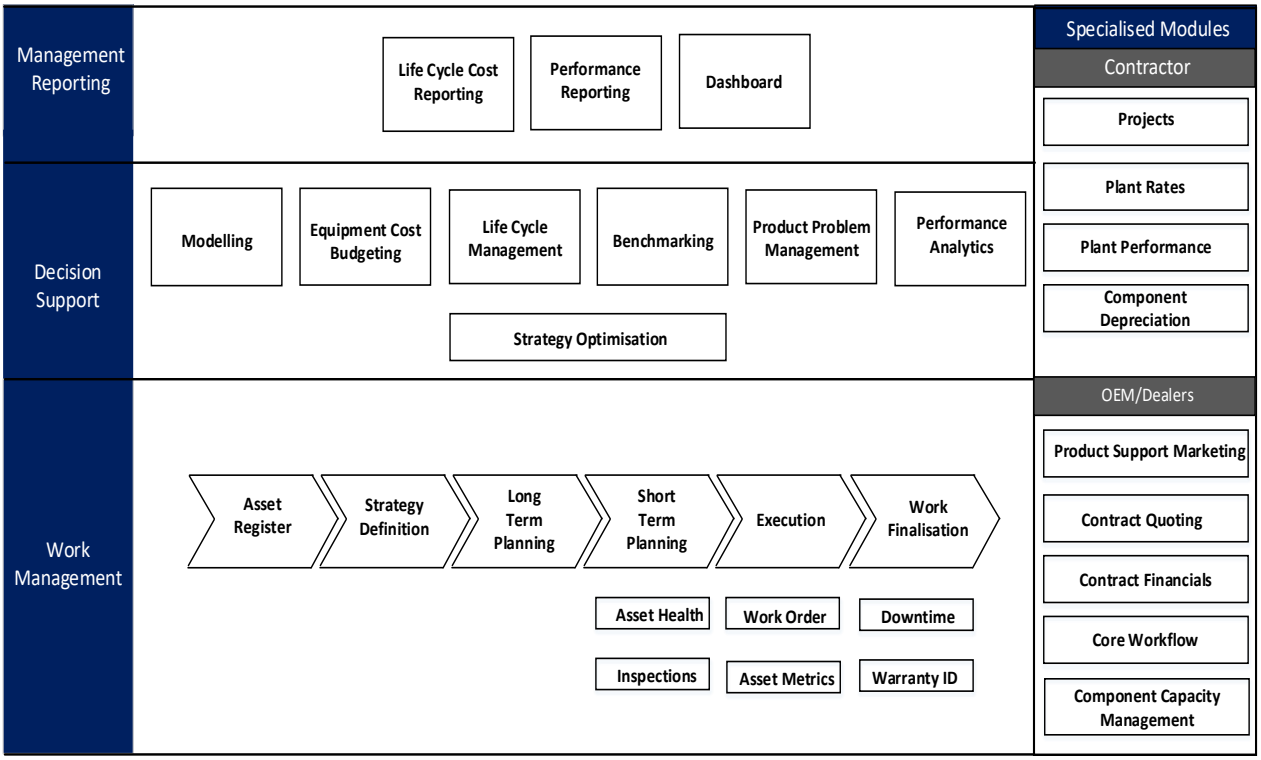
Check your understanding on the workshop and ask any related questions.

Course Outline

Pre-requisites	Getting Around AMT, AMT Key Concepts
Target Audience	AMT Administrator / Champion
Nominal Duration	
Course Objectives	
Learning Outcomes	At the end of the course, participants will have learned to: • Authorise AMT users. • Administer system settings. • Maintain adequate security measures for the AMT system settings. • Maintain the integrity of all AMT system tables under the system administration menu list.
Assessment Strategy	The completion of workshop tasks
AMT Version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below.



Training Workshops

1. AMT System Administrator Guidelines

Overview – Guidelines

RESPONSIBILITIES

- To ensure properly trained and authorised staff are appointed as AMT administrator.
- To ensure that AMT users are properly authorised and licensed.
- To ensure that system administration settings are modified in a timely and appropriate manner.
- To ensure adequate security measures are implemented and maintained for the AMT system settings.
- To provide an agreed process for staff who want to change AMT system settings.
- To ensure changes to AMT system settings are communicated to appropriate personnel within the organisation.
- To ensure the AMT database is correctly administered.
- To ensure licensing agreements are in place before adding new users and machines to AMT.
- Maintain the integrity of all AMT system tables under the system administration menu.

PROCEDURAL REQUIREMENTS

- The Database Administrator approves all changes to the operation of the AMT server or SQL server databases.
- Connect new users to AMT only if they are covered by the licence agreement.
- The AMT Administrator approves any changes to system settings.
- Consider the consequences of a proposed change on users other than those requesting the change before it is implemented.
- Tell affected users about changes to settings before they are made, or as soon as possible after they are made.

AMT SYSTEM SETTINGS

- System administrators should verify the correctness of the initial AMT system settings.
- System administrators should periodically check the AMT system settings to ensure that they comply with your organisation's policy and they have not been altered without the knowledge of the system administrator.
- System administrators should keep relevant support procedures up-to-date and communicate changes to procedures to all AMT users.

TYPICAL AMT SYSTEM ADMINISTRATION TASKS

- Keep a register of all AMT client installations.
- Ensure that the number of AMT users complies with your current AMT licensing agreement.
- Keep this procedure up-to-date and communicate any changes to it to all AMT users.
- Ensure all changes to the AMT system settings are in accordance with company support procedures.

2. Software Configuration

Functionality Overview

One output of the AMT deployment process is the configuration of the AMT software. The configuration settings are sorted into the following general categories:

- Integration: interfaces enabled, file paths, file types and configuration.
- Modules: features enabled and configuration.

System configuration settings are not editable. The configuration settings are reviewed if:

- AMT is upgraded, and the new version has new configuration options.
- Additional interfaces are deployed.
- Additional modules are deployed.

3. User Profiles

Functionality Overview

Whenever a new user needs to be added or an existing user permission needs to be modified, the System Administrator needs to maintain groups, user names, passwords and group permissions.

USER ACCOUNTS

A user account links a user with a security group. A user must belong to a specified security group. The permissions for each group are set in group permissions (see the next section). To open this window, search and select the function **User Accounts**.

A user account requires the following settings:

- Login identification name (username)
- Security group, which assigns the user's level of access
- Interface language
- Full name
- Email address
- Restriction for which branches or sites the user can access
- Workorder Authorisation amount (if applicable)
- Quote Authorisation (if applicable) enabled, Maximum Authority and Maximum Discount
- Budget Restrictions and Status Permissions
- Attached Documents



*For Windows authentication, in the **User Account Details** form in the Login name field, enter the username in the format domain\username (for example, RPM\Darryl.Smith). For the AMT web application, switch off **Anonymous Authentication** in IIS for all AMT applications and switch on **Integrated Windows authentication**.*

GROUP ADMINISTRATION

This feature applies AMT access permission to each security group. Define the list of available groups and the type of access each group has to specific areas within AMT before setting the permissions and users. To manage user groups, search and select the function **Group Administration**.

CHANGE PASSWORD

Users can change their password by clicking **Change Password** in the login window or searching and selecting the function **Change Password**.

AMT does not place any restriction on the length, case or type of password that can be entered.

4. System Tables

Functionality Overview

The purpose of a system table and its contents are defined during the initial setup and installation of AMT. From that point forward, the AMT System Administrator is typically the only person who has access to directly modify the tables.

Interfaces from external systems to AMT can indirectly modify AMT system tables. For example, if AMT receives an equipment interface file that contains a region value (which does not already exist in AMT's Region system table), the interface process adds the new region to the system table to allow the interface to run successfully. The system tables that don't do this are the ones whose uniqueness depends on another field. For example, AMT cannot add a new work location by an interface because a work location must be associated to a site. The only exception to this rule is the Component Code Hierarchy; new codes are added to the subsystem that is flagged as the default.



Next Gen / EPF adapters will not allow system table data to be updated if it does not already exist.



The system administrator can indicate which option of some system tables is the default, using a Default Value flag. In normal operation, the user can make different selections if the default is not applicable.



A Budget or Budget Code flag is used in some system tables to define which codes can be used in strategy tasks (for example, which component codes can be used in the creation of a strategy task).

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

5. Tables – Organisation

Functionality Overview

Table	Description	Example	Relevance	Special Fields
Company Editor	Name of company	Newmont	Only relevant for reporting/filtering purposes	
Region	List of the regions that equipment can be assigned to.	APAC – Asia Pacific.	Only relevant for reporting/filtering purposes.	
Division	List of divisions within a company structure.	Heavy Construction.	Only relevant for reporting/filtering purposes.	
Branch Administration	Contains the branches that exist in AMT.	Branches within the company. This table can also represent different locations of equipment if all equipment are on the one site (that is, AMT is installed on one site).	Branch is one element of the equipment hierarchy (**Branch > Site > Fleet > Equipment**), and a branch can be linked to many sites.	
Site Administration	Contains the sites that exist in AMT.	Sites within the branch. This table can also represent different locations of equipment if all equipment are in the one branch (that is, AMT is installed in one branch).	Site is one element of the equipment hierarchy (**Branch > Site > Fleet > Equipment**), and a site can be linked to many fleets.	• Resource Scheduling options. • Shift Start Time and Week Start Day (for shift reporting). • **Currency**: for site level reporting. • **Branch**: A site is specific to a branch in AMT.

Table	Description	Example	Relevance	Special Fields
Fleet Administration	A collection of equipment per site, which is typically determined by reporting requirements.	If a mine site has trucks and excavators, all the trucks might be grouped under a fleet called “Haul Trucks” and the excavators under a fleet called “Hydraulic Shovels”.	Fleet is one element of the equipment hierarchy (**Branch > Site > Fleet > Equipment**). A fleet can only be linked to one site, but it can be linked to many equipment.	• Contains details for Service Accuracy reporting and Availability Forecasting (if enabled). • **Site**: A fleet is specific to a site in AMT.
Business Group	The part of your business where the equipment is being used. This table is flexible to meet your requirements.	For a mining customer, the business groups might be Drilling, Hauling, Construction.	This table should inter-relate to the Industry and Industry Group tables.	
Industry	The type of industry in which the asset operates. This table is flexible for to meet your requirements.	Mining, Oil and Gas, Forestry.	Required for identifying different industry groups.	
Industry Group	Subsets of an industry.	If the industry is mining, industry groups could include coal mining and gold mining.	At a high-level, this classification allows the user to run certain reports to gauge how assets perform in different applications (identified by different industries).	

Table	Description	Example	Relevance	Special Fields
Work Location	Locations at which a downtime event can occur.	Workshop bay 1, Field, Pit, Fuel bay.	By nominating where downtime events will take place, the user can run reports to see if more than one event is planned to occur in the same location, as well as inform staff where they need to go to perform the jobs.	• **Site**: A work location is specific to a site in AMT.
Work Group	Contains work groups that may be responsible for executing a task.	Field service crew, welding crew.	From a planning perspective, the user can nominate the number of available working hours per week per work group (not per employee). The user can also assign certain work groups to the execution of tasks.	• **Default**: Each site must have only one default work group, otherwise AMT will not successfully create work orders in the Planning module from long-term planning. • **Site**: A work group is specific to a site in AMT. If work group spans more than one AMT site, the same work group can be created for each site. • **Expired**: If a work group is no longer required, flag the work group as “expired”. Historical records will remain in the database for reporting purposes. • **Target Allocation%**: Used to assess whether work group resources are over or under allocated in work plans generated in the Planning module. Typically, this parameter is set to 80%.

Table	Description	Example	Relevance	Special Fields
Employee	List of employees. Employee information is primarily used in Resource Planning and Timesheet functionality, and labour resource reporting. The determination of whether a site uses work group resource planning or employee resource planning is configured at the site level.	Joe Bloggs, A5146.	From a planning perspective, if using employee level resourcing, the user can nominate the start and finish times per employee. The user can also assign certain employees to the execution of tasks.	• **Site**: Each employee can only be assigned to one site. • **Employee Number**: An employee must be unique and is usually aligned with the company's ERP/HR systems. If an employee works at more than one AMT site, the same employee needs to be created for each site with a new employee number. • **Terminated**: If an employee is no longer employed, flag the employee as terminated. Historical records will remain in the database for reporting purposes. • **Default Fields**: Can set work group, labour activity, secondary labour activity, sales groups, as required.

6. Tables – Financial

Functionality Overview

Table	Description	Example	Relevance	Special Fields
Cost Centre	Cost centres group costs for financial reporting purposes. Usually, the cost centre reflects the organisational or departmental structure of the company, and exists in the company's accounting system.	Hauling, Carbon Handling.	Cost centre is one of the key codes used to produce and analyse accounting budgets. Your company's finance group should define these codes.	• **CBD**: No longer used. • **CC Activity**: The cost centre's activity code and description, which is used for reporting in the Budget and Forecasting module. • **CC Location**: The cost centre's location code and description, which is used for reporting in the Budget and Forecasting module. • **CC Responsibility**: The cost centre's responsibility code and description, which is used for reporting in the Budget and Forecasting module.

Table	Description	Example	Relevance	Special Fields
Expense Element	A system table used to group costs for financial reporting purposes. Usually, expense elements reflect different categories of costs with the company's accounting system.	Tooling, Emulsion, Steel, Fuel.	Expense element is one of the key codes used to produce and analyse accounting budgets. Your company's finance group should define these codes.	• **Depreciation**: Used to indicate which expense elements are used for parent asset (that is, equipment) depreciation, which is used by the Asset Register module of AMT. • **Ownership Cost**: Used to indicate which expense elements are used for parent asset ownership costs (for example, insurance, finance, registration), which is used by the Asset Register module of AMT. • **CBD**: Used to indicate which expense elements are used for component-based depreciation on child assets (that is, components), which is used by the Asset Register module of AMT. • **Global Code**: Links to the Global Expense Element system table.
Cost Responsibility	A high-level system table that defines the responsibility for equipment.	Site A Production.	Often, cost responsibility is a higher grouping of cost centres. In AMT, it is used primarily for financial and budget reporting.	

<table>
<tr><th>Table</th><th>Description</th><th>Example</th><th>Relevance</th><th>Special Fields</th></tr>
<tr><td rowspan="3">Currency</td><td rowspan="3">Lists all the currencies that could be used by an organisation for transactions. These currencies are set up in AMT during installation and should only be changed in consultation with RPM. The user enters a symbol to denote the currency (such as $) and default labour rate per hour for that currency.</td><td rowspan="3">AUD – Australian Dollars – $.</td><td>AMT can process multiple currencies because some organisations incur costs and make sales in different currencies.</td><td>• Default Labour Rate: The baseline labour rate that is applied to the Labour Rate system table when new branches are created in AMT.</td></tr>
<tr><td colspan="2"> 
The first currency in the currency table is flagged as the primary currency for the database. All other currencies are converted using exchange rates to the primary currency for reporting purposes. After the primary currency is set, it cannot be changed.</td></tr>
<tr><td colspan="2"> 
Deleting an unused currency causes all related exchange rate data to be deleted.</td></tr>
<tr><td>Entry Distribution Code</td><td>Distribution codes are the codes that allow the interfaces to post plant hire costs to the ERP system general ledger of the company using a journal transaction.</td><td>101 – Initial Purchase – Total asset cost</td><td>Entry Distribution Codes are used to identify the balanced transactional journal that is created by AMT when an asset process is executed. The account codes associated with the debit and credit detail will align to the ERP financial ledger codes (representing Balance Sheet and P&L accounts) and support any interfaces developed to post transactions to the ERP general ledger.</td><td></td></tr>
</table>

Table	Description	Example	Relevance	Special Fields
Escalation	Named Escalation Indexes can be created and the historical escalation factors can be entered. These indexes can then be associated to the parts, labour or miscellaneous costs for any projected task.	PartsEsc – Escalation for Parts Cost	Escalation refers to price changes over time and is sometimes referred to as indexation. Named escalation indexes can be created in AMT. The escalation factor at different time points can be entered for these indexes. These escalation indexes can be associated to the parts, labour or miscellaneous costs for any projected task. The current cost can then show the correctly indexed cost, and historical work order settlements can be analysed using the cost escalated to today's index.	• **Parts Default**: Default escalation for parts. • **Labour Default**: Default escalation for labour. • **Misc Default**: Default escalation for miscellaneous.
Exchange Rate	The exchange rate of a nominated currency expressed as one unit of the primary currency (see Tables > Financial > Currency).	NZD / AUD = 1.13 (where AUD is the primary currency).	AMT forecasts future costs and sales, which come from projections (typically current projections). The costs and sales in projections can be set up with different currencies; however, these transactions must be converted to be reported (generally) in the equipment's site currency. The exchange rate tables provide AMT with the current exchange rates at which to convert the costs or sales in projections to the primary currency to report future expected costs and sales.	

Table	Description	Example	Relevance	Special Fields
Financial Periods	The company's accounting periods. If the accounting periods differ from calendar periods, a translation process converts financial year and month records to calendar year and month, in the format YYYYMM.	If the organisation's financial year ends in June 2023 and an accounting record for a work order shows the invoice period is 202308, this will translate to February 2023 in calendar years 202302).	Certain financial reports require the user to select the accounting period rather than the calendar period.	 *Do not change this system table without consulting RPM.*
User Defined Financial Periods	The company's definition of financial periods for depreciation, identifying start and end dates for each period.	This sets the current period for depreciation calculation. Asset-related journals are posted to this period.		 *Do not change this system table without consulting RPM.*
Reporting Calendar	System table to define custom reporting calendars.			
Overhead Cost Categories	Categories for the different types of overhead costs, which can be related to a specific combination of cost centre and expense element.	Mining Maintenance (overhead cost category) - Hauling (cost centre) - Tooling (expense element).	The categories in this table reflect the type of overheads to be captured and reported.	• **Cost of Sales**: Typically, overheads are classified as operational expenses. If the company categorises certain overheads as costs of sale, those overhead categories should be flagged as Cost of Sale.

Table	Description	Example	Relevance	Special Fields
Set Margin	System table to set default margin percentages, for the automatic calculation of future cost at cost. Parts margins can be set per system. Margins can be adjusted per fleet if required.		Forecasted Cost at Cost in AMT is calculated using the Costs at Sell figures stored in a projection per task LESS the margin for the fleet as set up in the **Set Margin** system table.	
Sales Status	Status values for sales.	Sale Confirmed.	The values in this table reflect the statuses of components to be captured and reported.	• **Default Record**: Default value for all components. • **Probability Percent**: Accuracy of the status.
Budget Type	System table that defines the budget types that are used in the budget header.	Annual Budget	Used for filtering and reporting in the Budget and Forecast manager.	
Budget Status	Validates entries in the Edit Status field of the Budget Header.	Authorised		

7. Tables – Maintenance

Functionality Overview

These tables are for registering maintenance data. All costs incurred by an asset are allocated to a component code. Accordingly, component codes are used to create projections and to identify on which part of an asset work was performed.

The component code hierarchy therefore establishes the relationship between different component codes, usually in terms of location on the asset. There are four levels to the component code hierarchy - cost type, system, subsystem and component code.

Table	Description	Example	Relevance	Special Fields
Cost Type	Cost types are the top level of an equipment hierarchy.	Repair & Maintenance, Capital, Operator.	The component code hierarchy tables also provide a structure for analysing data.	
System	Systems are the second level of an equipment hierarchy.	R&M * Engine, R&M * Drive Train.	The component code hierarchy tables also provide a structure for analysing data.	• **Cost Type**: Systems must be associated to a cost type.
Sub System	Subsystems are the third level of an equipment hierarchy.	R&M * Engine * Fuel System.	The component code hierarchy tables also provide a structure for analysing data.	• **Default for Unassigned**: When AMT receives new component codes through an interface, the codes are allocated to this default subsystem. • **Expense Element**: The default expense element assigned to the subsystem's unassigned task. • **Systems**: Subsystems must be associated to a system.

Table	Description	Example	Relevance	Special Fields
Component Code Hierarchy	The user assigns component codes to different subsystems.	R&M * Engine * Fuel System * A982 Fuel Tank.	The component code hierarchy tables also provide a structure for analysing data.	• **Budget Code**: Only component codes flagged as budget codes can be used in a strategy task. • **Global Component**: Links to the Global Component Code system table. • **Alert**: Flags when the component code description equals the component code, which is usually a result of AMT receiving a new code through an interface.
Job Code	A list of the types of jobs that can be performed when a task is executed.	J01 – Remove and Install, J02 – Replace with exchange.	A task may consist of one of more jobs, each of which can have a cost and an associated bill of materials. AMT usually replicates the job codes that already exist in the ERP system.	• **Budget Code**: Only job codes flagged as budget codes can be used in a strategy task. • **Global Job Code**: Links to the Global Job Code system table. • **Flag as a Lube**: Flags a job code as lube if it is to be used in a strategy task configured as a consumable. This enables the consumable strategy task to be forecast in the Consumable Forecast report. • **Consumable Description**: If the job code is flagged as Lube (consumable), enter the description of the consumable (such as engine oil).

Table	Description	Example	Relevance	Special Fields
Job Code (cont.)				• **Consumable UOM**: If the job code is flagged as Lube, enter the UOM of the consumable (such as litres). • **Remove and Install**: Flag the job code as the remove and install job code for it to be classified as that on the Agreement Summary 2 report. • **Include R&I Job**: Used in conjunction with the Remove and Install flag when creating strategy tasks from standard jobs with the same combination of CC, MC, TT and TC. • **Shop Labour**: If the job code is flagged as Shop Labour, a strategy task that is created from a standard job with this job code will be set as Shop Labour.
Modifier Code	List of codes used to provide an additional identification to the component code, such as its position on the asset.	R – Right, L – Left, RF – Right Front, T – Top.	AMT usually replicates the position codes that already exist in the ERP system.	• **Budget Code**: Only modifier codes flagged as budget codes can be used in a strategy task.

Table	Description	Example	Relevance	Special Fields
Occurrence Type	A list of failure modes that can be selected by the user when recording the nature of a maintenance task performed.	AF After Failure, BF Before Failure.	Used by the technical analysis reports to highlight the nature of the maintenance tasks performed, for example, the number of components changed out after failure versus before failure.	
Task Counter	Used to distinguish between similar strategy tasks. The task counter field either remains blank or is used to distinguish between strategy tasks that have the same combination of component code + modifier code + task type.	1, 2, 3.	If you are mining company setting up an equipment under a MARC, the task counter can used to distinguish between different billing steps.	• **Budget Code**: only modifier codes flagged as budget codes can be used in a strategy task.
Task Type	A code and description combination, which identifies the general activity to happen at an occurrence of a task.	CC – Component Changeout, IN – Inspection, GR – General Repair.	Task Type is a unique code to AMT. If a task type is used to identify PM services, this should be flagged as a PM planning task. Can also be replicated from the maintenance activity type in the ERP.	• **Maintenance Type**: Links to the Maintenance Type system table. • **PM Planning**: Task types flagged as PM will be included in service accuracy statistics. • **PM Task**: Task types flagged as PM will be included in Quick Watch's First Stoppage After PM statistic.

Table	Description	Example	Relevance	Special Fields
Task Type (cont'd)				• **Comp. Changeout**: determines which strategy tasks are presented on AMT Mobile's Customer app as opportunities. • **Rebuild**: AMT has a special cut-down view for "rebuild" work orders, and these work orders can capture extra information with respect to rebuild turn times. Rebuild work orders are used by those using AMT work orders to rebuild components. • **Inspection**: Must be flagged to perform a scheduled inspection for the task type in AMT Mobile.
Maintenance Type	A high-level maintenance code used for reporting.	Shop, Field.	A high-level maintenance code used for reporting.	
Maintenance Order Type	Maintenance code used for reporting.	Corrective Maintenance	This code is not dependent of AMT Task Type. For AMT4SAP solutions, these codes can mirror SAP Maintenance Order Types if reporting by these codes are required. Or these codes could simply be Preventative Maintenance (PM), Corrective Maintenance (CM) and Breakdown Maintenance (BD).	

Table	Description	Example	Relevance	Special Fields
Activity	Describes the type of activity that can occur during a downtime event.	Perform maintenance, wash machine, waiting parts.	In Equipment Status, users can select different activities in the downtime allocation section of an event to describe what happened during a certain period. For each activity there must be a variance cause (see the Variance Cause table).	
Cause	List possible causes of the occurrence of a task.	Fatigue, Cracked, Over pressure.	Mechanism supporting reliability centred maintenance, through which users can analyse the causes of failure.	• **Global Cause**: Links to the Global Cause Code system table.
Delay Code	Code that identifies the reason for a work order being delayed.			
Event Prevention	List of proactive methods that could be used to prevent breakdowns.	Oil analysis, inspection, operator training.	If an event is classified as a breakdown, the user must nominate how the breakdown could have been prevented by selecting an option from the Event Prevention table.	
Source	Refers to the source of maintenance information when creating a downtime task, for example, reported by the operator, daily inspection, oil sample analysis.	Inspection, application study, operator comments.	Provides an audit trail back to the originator of a new task in case more information is required.	• **EMI Action**: The source assigned to work orders generated through the EM Issues module. • **Customer Reported**: The Product Incident Report Capture functionality requires a default source to reflect work orders raised through this process.

Table	Description	Example	Relevance	Special Fields
Symptom	List possible symptoms of a task.	Leaking oil, low power, and slow hoist.	Mechanism supporting reliability centred maintenance, through which users can analyse the symptoms of failure.	• **Strategy Default**: The default symptom that should be assigned to work orders raised by strategy tasks.
Task Planned Time	Options available to the user for identifying when a task could be planned for in the future.	Next operational stoppage, next planned stoppage.	Provides guidance to a planner as to when a task should be scheduled into a work plan.	• **Plan Days Offset**: When a backlog work order is assigned, a task planned time, the work order's target date will be calculated as today's date + plan days offset.
Variance Cause	Contains reasons for a variance between planned downtime and actual downtime.	Insufficient parts in stock.	Used by maintenance groups to identify inefficiencies in maintenance processes.	• **Activity**: For each activity (Activity system table) there must be at least one variance cause, and the description of the variance cause should be relevant to the activity. For example, if the activity is "waiting parts", a variance cause might be "insufficient parts in stock" or "parts not supplied in advance".
Issue Type	Classifications for an EM Issue.	WHS – Workplace health and safety, PP – Product problem.	Allows the user to run reports by EM Issue type, because some issues may require more urgent attention than others (such as safety issues).	
Workorder Settlement Status	Designates the status of a work order.	REL - open, CLSD – closed, IP - in progress, TECO - technically complete.	Required by AMT's Workorder interface to determine the status of work orders imported from external systems.	

Table	Description	Example	Relevance	Special Fields
Supplier	Lists suppliers of parts and miscellaneous resources (for example, subcontractor labour hire company).	Ford, subcontractor labour hire company, Hitachi.	Assigning suppliers to parts and miscellaneous items on a work order is required to generate purchase orders in AMT. Purchase orders can be emailed, printed or in certain AMT installations interfaced to external systems.	
Scheduling Group	List of scheduling groups used for suppression and dependency logic.	Suppression A, Dependency X.	Tasks can be assigned to scheduling groups. These groups can be performed either cyclically using dependency or in a parent/child relationship using suppression.	• Refer to the Scheduling Methods training and help for details.
Task Misc. Category	High-level categories of miscellaneous maintenance resources.	Subcontract Labour, External Equipment Hire, Safety.	Miscellaneous resources used on AMT work orders and standard jobs can be grouped by Task Misc. Category.	
OH&S	Occupational Health and Safety attributes of a work order.	Statutory, Environmental, Safety.	Used to describe the OH&S aspects of an AMT work order.	• **Auditable Record**: If the user selects the OH&S option, which is flagged as the Auditable Record, in an AMT work order, certain fields in the AMT work order are disabled. The specific fields disabled are Workorder Description, Component Code, Modifier Code, Task Type and Task Counter, Source, Raised By and Date Notified.

Table	Description	Example	Relevance	Special Fields
User Work Order Status	User work order status codes reflect the state of a work order from a user perspective not covered by the system status.	Pending Production, Awaiting Parts.	These codes are available to assist managing the status (or state) of a work order from a user perspective during the planning and scheduling lifecycle of a work order. For example, the user may want to know what work is pending production, pending scope, awaiting parts or awaiting external labour. Such states can be defined as a user work order status and applied to work orders. Work orders can be searched using this status as well. The user work order status cannot be changed on a work order with a system status of In Progress, Abandoned or Completed.	

Table	Description	Example	Relevance	Special Fields
Planning Group	Defines a person or group of people responsible for planning maintenance activities. You can belong to zero, one or many Planning Groups. Including a Message Group reference lets email notifications to be sent to the Planning Group.	Electrician Planning Group, Mobile Team 1		• Typically, the Planning Group Code will allow the easy identification of the person or team responsible for planning the maintenance activity. This is not a mandatory field but if populated will allow filtering and grouping of maintenance activities across various AMT screens. This filtering and grouping will be exposed via the Long Term Planning - Advanced feature. Membership Count identifies the number of active AMT Users that are part of this Planning Group.
Operational Criticality	Enables a risk rating or priority to be given to the maintenance activity. If the AMT Risk Feature is enabled, ranking supports further calculations of overall risk in long term planning and budgeting. Operational Criticality can be used without the Risk Feature enabled but will not support risk calculations.	Urgent, Low	Operational Criticality allows a single code to be identified as the default. Each code can have a ranking of 1 to 5 defined. Ideally, the same ranking would not be used against multiple codes, however, to support upgrading customers who have defined this field prior to the Risk Feature, and who have greater than 5 existing codes, AMT will support the same ranking against different codes within this table.	

Table	Description	Example	Relevance	Special Fields
Notification Type	Used to identify different reasons for creating a notification. Different types of notifications may be addressed differently from a work management perspective. A notification type supports the grouping of notifications.	Maintenance Request, Activity Report		
Operation Work Status	Used to identify different statuses for a work operation. Operation Work Status allows mobile users to redirect work order operations and alert supervisors and next shift work groups of a work order operation's progress.	Abandoned, In Progress	A Work Order Operation can have one and only one status at any point in time. The Work Order Operation has an additional flag that indicates it completion. That flag is independent of this status. This status has been introduced to support AMT Mobile processes which occur more naturally at a work order operation level and no longer includes paper documentation that indicates progress/completion status.	

Table	Description	Example	Relevance	Special Fields
Regulatory Code	Used to identify different legislative or regulatory compliance applicable to a work order or work order operation. One and only one code can be held against a work order header or work order operation. The work operation Operation code can be different from that at the header.	Mining and Quarrying Safety & Health (Canada), Mines Department (Qld)	The Regulatory Code can be defined to suit reporting requirements. It could be a simple yes/no option or it can be more detailed and identify the different compliance reporting bodies. If identification is needed to cover multiple legislative bodies, a code will need to be defined to cover this.	
Source of Supply	Used to identify the supplier/origin of a part. A part is uniquely identified in AMT by the combination of part number and source of supply.			• The Code indicates the code of the supplier from whom the part was purchased.

Table	Description	Example	Relevance	Special Fields
Object Part	Used to identify an asset part impacted when recording failure assessments.	Axles, Drive Shaft	The Object Part may be synonymous with a component code or may be a different level of detail (greater or lesser) depending on any FEMA or FEMCA reporting being undertaken. Subject to enabling the AMT feature Catalogue Profile, the object part is linked with an Equipment Catalogue code to provide a selectable list of asset parts relevant to that equipment catalogue.	

8. Tables – Parts & Standard Jobs

Functionality Overview

Table	Description	Example	Relevance	Special Fields
Labour Activity	A description of types of labour.	Auto Electrician, Mechanic, Welder.	Used for forecasting (strategy task) and planning (work order) labour requirements in terms of skills required. Standard jobs and/or work orders may require one or more different labour activities, each of which can have a specific labour rate.	
Labour Rate	A combination of labour activity, branch and currency for which a labour rate per hour can be assigned.		Labour rates apply to standard jobs or work orders that include labour activities. When a labour rate is changed in the Labour Rate table, all priced standard jobs that are "Calculated from Operations" and for which there is an operation using the changed labour rate are updated. There can only be 1 labour rate for every combination of branch, currency, labour activity, pay type.	

Table	Description	Example	Relevance	Special Fields
Part Type	Groupings of parts such as exchange, new, reman, fluid.	Reman, General.	It may be necessary to set up different categories/types of parts for reporting purposes. Part Type can be used in a strategy task's job to identify whether the job's costs are based on a specific part. For example, if the costing of a strategy task's job is based on using a reman component, then part type might be Reman, and if the costing of a strategy task's job is for a consumable (such as fuel), then part type might be Fluid.	• **Global Code**: Not relevant. • **Fluid Type**: Not relevant.
Parts Price Group	Table for registering the manufacturers of equipment or assets.	REN – Renault.	Manufacturers provide a basis for registering different models of equipment and for escalating parts costs in AMT.	• **Export Code**: Not used.

Table	Description	Example	Relevance	Special Fields
Parts Manager	Table for registering models of equipment/assets produced by manufacturers.	Renault – RSC182.	Every asset registered in AMT must be linked to a model. When the manufacturer of a model is changed, AMT checks to ensure the relationship between the serial number of the asset and manufacturer are not duplicated. For example, two current assets have the same serial number (SN 01HL00003 for manufacturer XYZ, model 123 and SN 01HL00003 for manufacturer ABC, model 987. The user cannot change manufacturer ABC for model 987 into manufacturer XYZ, otherwise the serial number 01HL00003 will not be unique for the manufacturer XYZ).	• **Manufacturer**: A model must be associated to one manufacturer. • **Equipment Group**: A model must be linked to one equipment group. • **Expense Element**: Default expense element for parts, labour and miscellaneous. This expense element will be used for Standard Job Cost Allocation if the part, labour or miscellaneous item included on the Standard Job does not have a uniquely identified expense element.
Part Classification	This is a flexible system table used for financial reporting.	Truck, Excavator.	Organisations may use different codes for different purposes.	
Parts Warranty Group	Defines the warranty period for parts.	GEN – General Parts, 180 Days.	Used when importing parts usage details to determine if there is a potential warranty claim for that part.	
Part Rating	Designed to capture a comparison attribute of the component.	250kw, 3600hp.	Used in forecasting and analysing components.	

Table	Description	Example	Relevance	Special Fields
Inventory Parts	The Inventory Parts function is used to manage the status of components in inventory and/or on order. This can be imported or exported.		Used in component capacity management and rebuild centre workload functionality and reporting.	
Commodity Code	A two level grouping for parts reporting.	Brake parts, fluids.	A reporting code used in AMT's Product Support Marketing module by OEMs and dealers.	
BEC Code	This table allows you to create Broad Economic Category (BEC) Codes.	Oil, Engines, Track Group.	These codes group transportable goods according to their main end use. This code is only relevant for the Product Support Marketing module.	
PACC	This is a flexible field, which can be used to group strategy tasks for reporting and analysis by AMT's Product Support Marketing module.		This code is only relevant for the Product Support Marketing module.	

Table	Description	Example	Relevance	Special Fields
Delivery Method	Identifies the method by which a purchase order or issue requisition parts can be nominated by the user to be delivered. It is applied to a purchase order or issue requisition and identifies how parts are to be delivered. It can be passed via integration to procurement solutions.		Only one delivery method can be nominated per purchase order line item. A default delivery method can be nominated and if so, will default to each line item where it can be overwritten, Delivery method can be modified by a user using Edit Work Order. Delivery method is not mandatory.	

9. Tables – Equipment

Functionality Overview

Table	Description	Example	Relevance	Special Fields
Equipment Class	High-level grouping of the different equipment types.	Mobile Equipment, Fixed Plant, Transport, Infrastructure.	Filter available in high-level financial reports.	
Equipment Criticality	Groups to represent how important an equipment is to the organisation, operation, site, or profitability.	Overburden, Hauling, Drill & Blast, Power Generation.	Filter available in high-level financial reports.	
Equipment Group	Provides a group of the model codes. Equipment group is sometimes referred to as model family.	OHT - Off Highway Truck.	Within AMT's asset register, the technical details relate to the equipment group (that is, each equipment group will have its own set of technical details).	• **Include Tech Fields**: Different equipment registration, configuration and application data fields can be enabled here. Enabled options appear in the Equipment Plan Editor screen for the user to populate. Options highlighted in red cannot be deselected because users have already entered data into these fields within equipment.
Manufacturer	Table for registering the manufacturers of equipment or assets.	REN – Renault.	Manufacturers provide a basis for registering different models of equipment and for escalating parts costs in AMT.	• **Export Code**: Not used.

Table	Description	Example	Relevance	Special Fields
Model	Table for registering models of equipment/assets produced by manufacturers.	Renault – RSC182.	Relevance: Every asset registered in AMT must be linked to a model. When the manufacturer of a model is changed, AMT checks to ensure the relationship between the serial number of the asset and manufacturer are not duplicated. For example, two current assets have the same serial number (SN 01HL00003 for manufacturer XYZ, model 123 and SN 01HL00003 for manufacturer ABC, model 987. The user cannot change manufacturer ABC for model 987 into manufacturer XYZ, otherwise the serial number 01HL00003 will not be unique for the manufacturer XYZ).	• **Manufacturer**: A model must be associated to one manufacturer. • **Equipment Group**: A model must be linked to one equipment group.
Equipment Category	This is a flexible system table used for financial reporting.	Truck, Excavator.	Organisations may use different codes for different purposes.	
Equipment Location	These are general categories describing locations of equipment.	Locations could be production areas (such as Pit A), but should not replicate regions, branches, or sites.	This flexible system table can be used by organisations for different purposes. This code is not linked to the equipment hierarchy (**Branch > Site > Fleet**).	

Table	Description	Example	Relevance	Special Fields
Asset Status	Asset Status value for equipment.	Parked Up, Off-hired.	This flexible system table can be used by organisations for different purposes. This code is not linked to the equipment hierarchy (**Branch > Site > Fleet**).	
Asset Condition	Used to identify the health of an asset. Asset Condition enables a risk rating to be given to the asset. If the AMT Risk Feature is enabled, ranking supports further calculations of overall risk in long term planning and budgeting. Asset Condition can be used without the Risk Feature enabled but will not support risk calculations	Very Poor, Good	Asset Condition allows a single code to be identified as the default. Each code can have a ranking defined; AMT will support a ranking from 1 to 10. The same ranking would not be used against different Asset Condition codes.	

10. Tables – Functional Location

Functionality Overview

Prior to starting the functional hierarchy process, RPM recommends you work out how you want to set up your structure, to reduce rework if you want to change something.

Table	Description
Functional Location Category	Add, edit and delete functional location categories.
Functional Location Level	Add, edit and delete functional location levels. These are used by AMT to designate the labels for the levels of the hierarchy. You can change the descriptions associated with each level.
Functional Location Attribute	Add, edit and delete functional location attributes. AMT can accommodate up to 30 attributes for a functional location. This table helps you ensure that any functional locations from your ERP system are replicated by AMT.

11. Tables – Usages & Measurements

Functionality Overview

These tables are for registering usage data.

Table	Description	Example	Relevance	Special Fields
Units of Measure	A defined unit of measure.	Days, hours, kilowatts, litres, tonnes.	Required to read measurements and create usage profiles.	• External UOM: Unit of measure code used by an external system that aligns to the UOM in AMT. • Operating Hours: For any equipment to show data in the **Scheduled Hours** form, its primary UOM must be flagged as Operating Hours. It is possible to flag multiple UOMs as Operating Hours. A UOM should be flagged as Operating Hours if it is to be used in availability, reliability and production statistic data analysis. • Revenue Rate: Not used.
Measurement Code	Codes for specific measurements.	Fuel quantity, oil level, engine temperature.	Used in consumption reports and in the Condition Based Monitoring module to define measurement translation rules and capture actual condition monitoring data.	• Workorder Measurement: Filter manually recorded measurements on the Measurement tab of the Workorder Feedback tab.

Table	Description	Example	Relevance	Special Fields
Production Statistics Code	Codes used to describe the different operational statuses of an equipment.	Idle, Productive, Under Maintenance.	Used in Production Statistics reports only.	• Downtime: The code flagged as downtime will inherit actual downtime from AMT events. • Site: Different sites can have different codes. • Primary UOM: Flag to indicate whether the values are the same as the equipment's primary UOM for reporting purposes.

12. Tables – Other

Functionality Overview

These tables are for registering usage data.

Table	Description	Example	Relevance	Special Fields
Contract Type	Use this table to differentiate different types of maintenance contracts.	Full maintenance contract, cost cap	Contract Type is only relevant for identifying to which Department a Maintenance Contract belongs.	
Customer	A listing of Customers.		Used to identify the Customer who is operating Equipment.	
Document Type	Categories for the type of documents which can be stored in the Library.	Form, Inspection sheet		• When users add a document to the Library, the user must nominate the Document Type.
Global Cause Code	See Cause under Maintenance System Tables.			
Global Component Code	See Global Component Code under Maintenance System Tables.			
Global Job Code	See Job Code under Maintenance System Tables.			

Table	Description	Example	Relevance	Special Fields
Global Part Type	See Part Type under Parts & Standard Jobs System Tables.			
Global Expense Element	See Expense Element under Financial System Tables			
Strategy Task Marketing Grouping	This table is part of product support marketing. The marketing grouping is an additional grouping for strategy tasks which may be used by the sales responsibility to simplify groupings. The market grouping can be maintained in individual strategy tasks or using the marketing and customer support editor.	Product Support		
Log List Editor	Use this table to edit the reason codes for logging changes. This table is used when you change the next due details for strategy tasks.			

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

13. Data Translation and Validation

Functionality Overview

Code translation in AMT is designed to allow work order settlement component code combinations to be automatically translated during the work order interface process. This facility is typically used to allow a wide variety of component code combinations within an ERP (transaction) system to be expressed as a condensed version within AMT.

In work order settlement code translation, a user can:

- Add sets of external transaction system codes (component code, modifier code, job code) and link them to associated sets of internal AMT codes (these are also referred to as external/internal code mappings), so that when the transaction system codes are imported into AMT, they are translated to the AMT codes. The AMT codes have an extra field, Occurrence Type.
- Remove existing sets of external/internal code mappings.
- Edit existing sets of external/internal code mappings.
- Set the order of translation if there are data conflicts.

In AMT each work order settlement contains two sets of defining codes (component code, modifier code, task type, task counter, job code and occurrence type). One set contains the values of the codes received from the external system (these values will be null for internal AMT work order settlements). The second set contains the translated set of codes. The codes received from the external system will be functionally ignored and are provided for information purposes; conversely, AMT logic that requires the value of the codes will use the translated values.

AMT cycles through the set of new or updated work order settlements, and for each work order settlement performs one of the following actions:

- If the work order settlement links to an AMT work order, the internal AMT codes of the work order settlement are set to those of the AMT work order (that is, component code, modifier code, task type and job code).
- If the external set of codes that define the work order settlement (component code, modifier code, task type, task counter and job code) match a set of codes (the five codes specified above plus occurrence type) specified in the list of code translation rules, the internal AMT codes of the work order settlement are set to the values specified in that code translation rule. Otherwise, the internal AMT codes of the work order settlement are set to the values of the external codes.

When a field in a code translation rule has a value of null, this will act as a wild card for that field. For example, if there is just one code translation rule defined, as shown below:

CC	MC	TT	AC	JC	OT		CC	MC	TT	AC	JC	OT
1000	-	PM	-	-	-	→	1000	-	CC	Pair	010	B/F

The following work order settlements are imported into AMT from an external system, with codes as follows:

Work Order Number	Component Code	Modifier Code	Task Type	Task Counter	Job Code	Occurrence Type
WOA123	1000		PM	1	010	REMAN
WOB156	1000					
WOC964	1052		PM			
WOD777	1000		PM			

Work Orders WOA and WOD would satisfy the rules, and hence have their codes translated to those specified in the rule (that is, WOA and WOD would end up with the same set of codes). However, WOB and WOC would not have their codes translated - WOB does not have a task type of PM and WOC does not have a component code of 1000. So, WOB and WOC would be added to AMT with no change to their codes.

ORDER OF PRIORITY

All code translations are assigned an order of priority, as defined in the Order field. If two or more work order settlements contain, for example, four out of the five available code combinations, the translation will be determined on the code priority setting, so that only the highest priority code translation rule is applied to any work order settlement.

SET GL RECONCILING ITEMS

This function is used to manually reconcile transactions in AMT to records imported through the general ledger interface, where the interface was unable to link the records automatically (auto-reconcile).

LEDGER MISSING EQUIPMENT

This report displays transaction records imported through the general ledger interface that do not link to AMT equipment.

AMT TO GL RECONCILIATION

This report displays the results of the automatic reconciliation process between AMT transactions and general ledger transactions imported from an external accounting system.

RECONCILING OPEN WORK ORDER SETTLEMENTS

This report displays transaction records imported through the general ledger interface that are for open work orders. This report only displays data if AMT is configured in WIP mode (this mode is normally used only by OEMs).

WORK ORDER SETTLEMENT TRANSACTIONS

This report lists all work order settlement transaction information:

- Child work orders
- Financial period
- Every combination of work group, cost centre, expense element and cost bearer



This report is useful when trying to perform a manual reconciliation between AMT and an external accounting system where the general ledger interface is not deployed.

14. Configuration

Functionality Overview

Only relevant functions are described below.

GROUP PERMISSIONS

To manage access permissions, search and select the function **Group Permissions**. Types of access include:

- All: read and write access.
- Edit: no longer used.
- View: read-only access (forms only).
- None: cannot access that area of the program.

COST ALLOCATION INTERFACE DETAILS

This function allows the user to nominate which elements of the job cost (the cost allocation) interface will be updated by external systems and which elements can be modified manually in AMT by users.



No changes should be made unless verified first by RPM to minimise potential interface issues.

EXTERNAL DOWNTIME EQUIPMENT

This system table is only relevant if AMT is configured to import downtime notifications from external systems (such as MineStar, Jigsaw, and Modular Mining).

This system table is used to map the equipment name used by the external system to the equipment name registered in AMT.

EXTERNAL DOWNTIME SITE

This system table is only relevant where AMT is configured to import downtime notifications from external systems (such as MineStar, Jigsaw, and Modular Mining).

This system table is used to map the site name used by the external system to the site name registered in AMT.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

15. AMT Utilities

Functionality Overview

REFRESH REFERENCE DATA

Click this function to have the system table cache refreshed on your local AMT client. The user may need to start this function if the system tables have changed while they are still running the AMT client. Otherwise, every time the user logs into AMT, AMT refreshes the reference data automatically.

USER GROUP FUNCTION MANAGER

This feature shows a list of all functions accessible by a user group and the layout of the functions in the **My Function** list. To manage group function hierarchies, search and select the function **User Group Function Manager**.

RECALCULATION MANAGER

This function is used to manually force the recalculation of selected projections.

Only current projections are recalculated automatically by AMT. All other projections must be manually started the user to recalculate - either by using Recalculation Manager, or when prompted by the system.

INTERFACE ADMINISTRATION

This function allows the user to manually load specific interface files into AMT by browsing to the location where the interface file has been saved. This process does not permit the user to disable existing automatic interfaces nor change the timing of when existing automatic interfaces run.

Interface	Relevance
Overheads Cost Import	Used to import overhead costs into AMT if these costs are not already loaded into AMT using work order settlements.
Overhead Billing Import	MARC module installations only.
Planned Downtime Import (Deliverability)	Deliverability module installations only.
Plant Hire Charges Import	Plant Hire module installations only.
Plant Hire Rates Import	Plant Hire module installations only.
Plant Asset Journal Import	Plant Hire module installations only.
Add/Update Parts	Used to update the price of parts, and add new parts to AMT, where AMT is used to define standard jobs.
Usage Profile Rates Import	Used to import projected usage rates into AMT for equipment linked to usage profiles.

Interface	Relevance
Electronic Workscope Import	Used to import workscopes into AMT.
Electronic Workscope Steps Import	Used to import workscope steps into AMT.
Serialised Component Import	Used to import serialised components into AMT.
Inventory Parts Import	Used to import inventory parts into AMT.

EXPORT ADMINISTRATION

This function is used to export specific data out of AMT in the format used by interfaces to load that same data into AMT. This function provides a quick way to extract, cleanse (modify) and reload data into AMT.

Interface	Description
Export Equipment	Export equipment registration details out of AMT.
Export Strategy Tasks	Export strategy task details out of AMT (does not export cost allocations or standard jobs linked to strategy tasks).
Export Utilisation	Export actual usage recordings.
Export Workorder Settlements	Export actual work order settlements.
Export Planned Downtime	Export planned downtime.

INTERFACE LOG

This function provides a report detailing the error log of the interface between AMT and the ERP for the following interface files:

- Import Equipment
- Import Strategy Tasks

An administrator can review this log to determine the root cause of why data was not inserted when the interface last ran.

GENERIC INTERFACE LOG

This report lists errors generated by any interface file imported into AMT. The report lists: filename, error description, the line of the interface that contains the error, and the date when the interface file was processed.

SYSTEM TASK MANAGER

This function provides a form for the administrator to view and create AMT internal system tasks, especially those encompassed by the AMT/ERP interface. Users can queue or run system tasks on demand (by right-clicking a task and selecting **Run Now**) and view when a system task last ran. The user is provided with details such as last run date and time, next scheduled run date and time and manually create a task to run. Only system tasks relevant to the setup of AMT will be enabled during system configuration by RPM.

Below is a list of some of the processes contained in the system task manager:

Process	Description
Recalc (Current)	Recalculates current projections only.
Recalc (Budget)	Recalculates budget projections only.
Replanning	Calculates incremental usage per UOM per equipment, identifies new overdue strategy tasks, and reschedules existing over dues to the next day.
Archive Projections	Creates archive projections when requested.
Process Interface files	Processes new interfaces from external systems.
Parts Forecast	Generates parts forecasting data (specific module).
KPI Calculation	Recalculates Dashboard statistics.
Update IP Events & Equipment	Updates the in progress events on the Equipment Down Gantt Chart in Equipment Status.
Set Dirty Billings	Refreshes billings (MARC only).
Email Queue Manager	Generates emails queued in AMT.
CBM Action Process	Executes CBM translation rules to determine whether new alerts should be posted.
Message AutoDelete	Delete old messages based on system configuration parameters (parameters defined when AMT was installed).
Create Snapshot	Process for generating snapshot data for specific AMT reports (for example, Change Analysis, Key Stats Change Trending).
Rebuild Index and Statistics	Process designed to optimise the performance of the AMT database.
Extract IC Data	Process designed to extract IC data sets (IC module specific).
Import IC Data	Process designed to import IC data sets (IC module specific).
SSRS Manager	Populates AMT with new reports developed in the IC module.

Process	Description
Import Cleanup	Deletes “staging tables” in AMT-IC (this data is kept for troubleshooting purposes, but data more than X days old is deleted).
Available Man Hours	For sites configured for employee level resource scheduling, this sums the available hours based on the employees for the workgroups.
Process SSAS	This process recalculates the SSAS cubes that are part of AMT IC module.
SRExport	Process to automatically export service reports from AMT that can be imported by external systems.
Delete Attachments	Process designed to delete documents that are no longer linked to AMT work orders (parameters defined when AMT was installed).
Update Planning from Strategy	Creates a work order in the AMT Planning module for the strategy task that has entered its planning window (reaches its Lead Days).
Parent Depreciation	Generates plant asset journals for parent asset depreciation costs from work order settlements.
Component Depreciation	Generates plant asset journals for components from plant hire invoices.
Update AMT Advanced Planning from WOS	Generates AMT events in the AMT Planning module based on an imported work order settlement (special configuration of AMT).
Create WO from WOS	Creates a work order in AMT’s Planning module based on an imported work order settlement (special configuration of AMT).
Update WO from WOS	Updates details of a work order in AMT’s Planning module based on an imported work order settlement (special configuration of AMT).
Mining Machine Down Alert – Excel	Process automatically generates an Excel file for the Mining Machine Down Alert report that is emailed to users (special configuration of AMT only).
WO Status Export	Generates work order status interface files which can be imported into external systems.

LANGUAGE TYPE

In AMT, you can redefine existing labels and captions. For example, changing the string “Component Code” to be “Machine Code”. This change would be seen on forms, screens and in reports.

The first step in this process is to create a new language, which can be an actual language (such as German or Japanese), or the language can be something like “Our Specific Labels”. The Language Type function is used to create the new language. Make changes to the labels and strings using the Language Editor function.

LANGUAGE EDITOR

This function is used to change or translate the labels in AMT for the language.



AMT does not translate data entered into AMT by users, only captions and labels. For example, the field label Component Code can be translated, but not the actual component codes entered into the component code system table. RPMGlobal does not maintain or update translations created by System Administrators. The default AMT language is English. The maximum length of all captions and labels in AMT is based on the English caption. AMT only supports the Gregorian calendar. AMT only supports left-to-right display of text.

CREATE WORKORDER SETTLEMENT HISTORY

This function is used to generate work order settlement history for strategy tasks flagged in this mode.

CREATE COMPONENT CHANGEOUT JOURNAL

This function is only relevant to Plant Hire installations of AMT.

This function is used to generate plant asset journals for component changeout work order settlements:

- **EDC 115**: Component Disposal.
- **EDC 116**: Disposal - Component - Reverse CBD.
- **EDC 110**: Component Capitalisation.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

16. Interface Process and Diagnosis

Functionality Overview

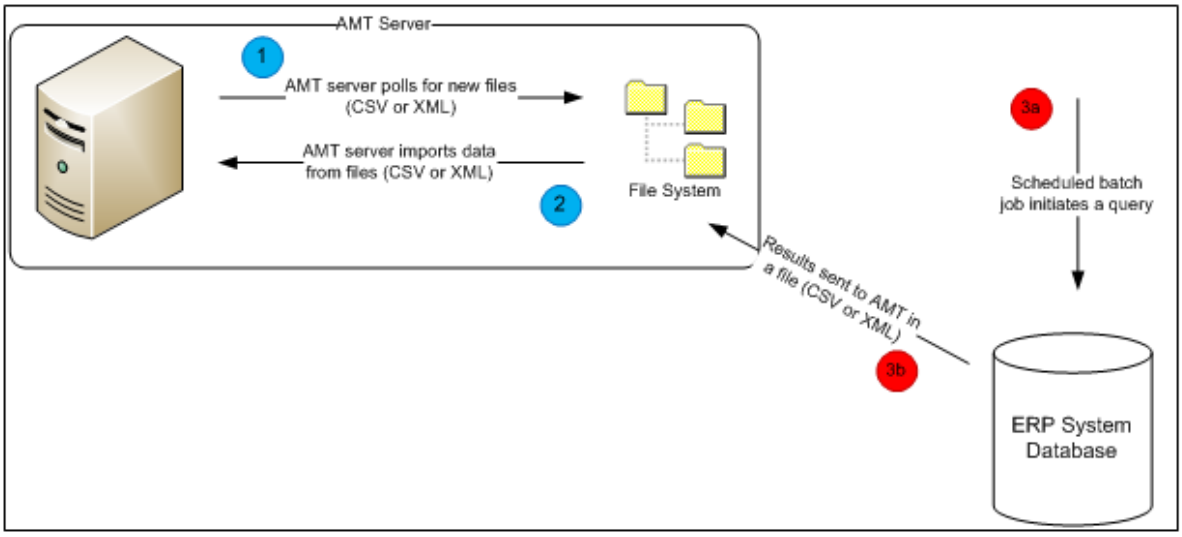
INTERFACE FILE TYPE

Typically interface files passed between AMT and an ERP are in CSV format; a file format that can be created and modified in Excel.

For CSV files, the first line contains the column names, and not all columns need to be supplied. Certain system integration messages do not support the use of CSV files, as they are hierarchical data structures (that is, more than one entity is contained in a message).

INTERFACE DATA TRANSFER PROCESS

Below is a diagram illustrating how interface data files are passed between AMT and an ERP.



On an AMT server, the AMT Calculation and Interface Windows Service application executes without user intervention. The service continually monitors certain file locations and imports any new files that appear in those locations. These interface files have a filename that includes an optional branch identifier and a time stamp. After files are processed, they are moved to a Success or Failure subdirectory.

INTERFACE DIAGNOSIS

Currently there is no mechanism in AMT to auto-generate interface errors. However, there are manual ways to check for errors. The following reports are provided in the AMT user interface:

- Generic Interface Log (refer to the AMT Utilities section of this document).
- Interface Log (refer to the AMT Utilities section of this document).

Using these reports and depending on the nature of the error reported, the AMT Administrator may need to review the contents of the actual interface file. This will require access to the AMT server and the ability to copy the interface file to their local computer. When reviewing the actual file, we strongly recommend the interface specification document is available to understand the rules and data mappings used to populate the file.

Typical data issues include:

- Mandatory interface file fields are missing data.
- Data inserted into the file contains additional commas, which can distort the separation of interface file fields. For example:
    - This problem is only encountered if the data in the interface file is not formatted with quotation marks; that is, oil, fuel and grease will cause a problem for CSV file processing (due to the insertion of commas) if it is not formatted like this with quotation marks “oil, fuel and grease”.
- Data is in the incorrect format. For example:
    - Date field set to 31-Jan-2021 instead of 20210131.
- Data flags or elements are not set up correctly in the ERP resulting in the wrong data being sent to AMT, data is missing, or data being sent correctly but not linking within AMT properly. For example:
    - The registration details of an asset might not be sent from the ERP to AMT if the asset is not flagged in the ERP as being required by AMT. Someone needs to set the flag against the asset in the ERP first.
    - Data for an asset might be sent to AMT (for example, strategy tasks, work order settlements) but since the asset is not registered in AMT, the associated data may not link to the asset properly (that is, work order settlements might link to overheads instead of linking to a proper asset).

There can be issues caused by business rule gaps, not handled by the interface’s programming logic, which causes expected behaviour in AMT. For example, a user may terminate an asset in the ERP but this may not terminate the asset in AMT. Business logic issues need to be worked through with the interface developer and potentially RPM to determine whether an interface logic change can solve the issue.



Not all business logic issues can be solved by interfaces. Where an interface can solve an issue, workarounds need to be defined, work procedures developed and users trained by the AMT Administrator.

INTERFACE REPROCESSING

After an interface issue has been resolved, files may need to be reprocessed into AMT. Reprocessing could involve restarting the interface between AMT and the ERP or there may be a need to reprocess a file manually.



If at all unsure about the correct approach to reprocess an interface file, contact RPM Customer Care for assistance. Care must be taken if manually reprocessing interface files especially when the file has been edited by Excel (for example, Excel drops leading zeros which could cause AMT to think a code is new and add it to a system table).

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes